Digital SAT Prep · Retired Paper SAT · No-Calculator Math

Paper SAT May 2023 — No-Calculator Math (Section 3) 20 Original SOMATH Problems

Original SOMATH problems modeled on the concept spread of the retired May 2023 US paper SAT No-Calculator section. Every question is written by the SOMATH team — not a reprint of College Board content. 25 minutes, no calculator. Answer key is on the last page. Full step-by-step solutions live at schoolofmath.us/posts/paper-sat-may-2023-no-calculator-math-20-original-problems

Question 1

A quiz has only 3-point questions and 5-point questions and is worth 60 points total. A scatterplot shows every valid combination as ordered pairs (number of 3-point questions, number of 5-point questions): (0, 12), (5, 9), (10, 6), (15, 3), (20, 0). If a student's plan includes exactly 5 of the 3-point questions, how many 5-point questions does the plan include?

A) 3
B) 6
C) 9
D) 12

Question 2

A linear function g passes through the points (0, −2) and (1, 1). Which equation defines g?

A) $g(x) = 3x - 2$
B) $g(x) = 3x + 2$
C) $g(x) = -3x - 2$
D) $g(x) = -3x + 2$

Question 3

The polynomial function h has x-intercepts at (−4, 0) and (7, 0). Which statement MUST be true?

A) $h(-4) = 0$
B) $h(7) = -4$
C) $h(0) = 7$
D) $h(-4) = 7$

Question 4

Solve the system: $y = 2x + 5$ and $3x - y = 4$. What is the solution (x, y)?

A) (−9, −13)
B) (−1, 3)
C) (1, 7)
D) (9, 23)

Question 5

What are all real solutions to $|x + 7| = 0$?

A) −7
B) 0
C) 7
D) −7 and 7

Question 6

The equation $p = t(w - 3)^2$ relates positive numbers p, t, and w. Which equation gives w in terms of p and t when $w > 3$?

- A) $w = 3 + \sqrt{(p / t)}$
- B) $w = 3 + \sqrt{p} / t$
- C) $w = -3 + \sqrt{(p / t)}$
- D) $w = -3 - \sqrt{(p / t)}$

Question 7

In a relationship between x and y, each increase of 1 in x decreases y by 4. When $x = 0$, $y = 9$. Which equation represents this relationship?

- A) $y = -4x + 9$
- B) $y = -4x - 9$
- C) $y = 4x + 9$
- D) $y = -(1/4)x + 9$

Question 8

A downward-opening parabola crosses the x-axis at $x = 0$ and $x = 10$ and reaches a maximum height of 25. It can be modeled by $y = -x(x - 10) / k$ for some constant k. What is the value of k ?

- A) 25
- B) 5
- C) 1
- D) 1/5

Question 9

An isosceles right triangle has a hypotenuse of length 6. What is its perimeter?

- A) $3\sqrt{2}$
- B) $6\sqrt{2}$
- C) $6 + 6\sqrt{2}$
- D) $6 + 12\sqrt{2}$

Question 10

How many solutions does the equation $5(x - 3) = 2x + 3(x - 4)$ have?

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

Question 11

Circle C has center (4, 3). Line m is tangent to circle C at the point A = (7, 7). What is the slope of line m ?

- A) $-3/4$
- B) $-4/3$
- C) $3/4$
- D) $4/3$

Question 12

The function $P(n) = 500 - 120(1.6)^{(-0.2n)}$ predicts a runner's finish time (seconds) after n weeks of training. Which expression represents the runner's predicted finish time after 0 weeks of training?

- A) P(−0.2)
- B) P(0)
- C) P(120)
- D) P(500)

Question 13

A machine's cycle time is 80 seconds on trial 1. On each following trial, the cycle time decreases by 25% from the previous trial. Approximately how many seconds is the cycle time on trial 4 ?

- A) 20
- B) 34
- C) 45
- D) 60

Question 14

Which table lists (x, y) pairs where every row satisfies BOTH $y < (1/3)x + 2$ AND $y > (1/2)x - 4$?

- A) (−3, −1), (0, 5), (6, 6)
- B) (−3, −6), (0, 1), (6, −1)
- C) (−3, 0), (0, 2), (6, −3)
- D) (−3, −2), (0, 0), (6, 0)

Question 15

Which of the following is equivalent to $\sqrt{50}$ · (cube root of 8) ?

- A) $10\sqrt{2}$
- B) 10 · (cube root of 4)
- C) $25\sqrt{2}$
- D) 100 · (cube root of 2)

Question 16

Line k passes through (0, 1) and (6, 3). Line j is parallel to line k. What is the slope of line j ?

- A) −3
- B) −1/3
- C) 1/3
- D) 3

Question 17

The expression $45x^8 / (9x^2)$ is equivalent to $c \cdot x^d$, where $x > 0$ and c and d are constants. What is c + d ?

- A) 5
- B) 6
- C) 11
- D) 30

Question 18

What value of h satisfies the equation $3.4(h + 2) = 4.4h + 3.4$?

- A) 1.7
- B) 3.4
- C) 6.8
- D) 10.2

Question 19

A cylinder and a sphere both have the same radius r, where $r > 0$. The cylinder has a height of 12. The volume of the sphere is one-third the volume of the cylinder. What is the value of r ?

- A) 2
- B) 3
- C) 4
- D) 6

Question 20

For the equation $x^2 + bx + c = 0$ with real constants b and c, the two roots satisfy $-b + \sqrt{(b^2 - 4c)} = 14$ and $-b - \sqrt{(b^2 - 4c)} = -2$. What is one possible value of x that solves $x^2 + bx + c = 0$?

- A) −7
- B) −1
- C) 1
- D) 7

Answer Key

Full step-by-step solutions are on the blog post:
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Q	Ans	Skill	Q	Ans	Skill
1	C	Read scatterplot	11	A	Tangent slope
2	A	Linear from graph	12	B	Function input
3	A	x-intercept ⇒ f(a)=0	13	B	Exponential decay 25%
4	D	Linear system	14	D	Two inequalities
5	A	|E|=0 special case	15	A	Radicals
6	A	Rearrange with root	16	C	Parallel-line slope
7	A	Linear from words	17	C	Monomial rational
8	C	Factored-form vertex	18	B	Solve linear
9	C	45-45-90 perimeter	19	B	Cylinder vs sphere
10	A	Count solutions	20	B or D	Sum/product of roots

About these problems

Every question in this worksheet is an original SOMATH problem written by our teaching team. They test the same underlying math skills the retired May 2023 US paper SAT No-Calculator section tested, but the wording, numbers, contexts, and answer choices are new. We do not reprint College Board content because it is copyrighted. The official released paper SATs are available directly from College Board's SAT practice archive at satsuite.collegeboard.org/sat/practice-preparation/practice-tests, and the current Digital SAT is at bluebook.collegeboard.org.